Technology Stack

Date	29 June 2026
Team ID	xxxxxx
Project Name	Voice Based Concept Understanding Analyser
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	**Frontend / Client-Side**(e.g., React, Angular, HTML/CSS)	**Streamlit, HTML, CSS**	Streamlit provides a responsive and interactive web interface for uploading audio, selecting concepts, viewing transcripts, evaluation metrics, waveform visualizations, and downloading reports with minimal frontend development.
2	**Backend / Server-Side**(e.g., Node.js, Python/Django, Java)	**Python**	Python serves as the core backend language, integrating speech transcription, semantic evaluation, audio feature extraction, scoring, and report generation into a modular application.
3	**Database / Data Storage**(e.g., MySQL, MongoDB,	**In-memory Processing (No Database)**	The current application processes uploaded audio files in memory without persistent

	PostgreSQL)		storage. Reference concepts are maintained within the application, making the system lightweight and easy to manage.
4	**Cloud / Hosting / Deployment**(e.g., AWS, Azure, Heroku, Docker)	**Local Streamlit Server** *(Deployment-ready for Streamlit Community Cloud, Render, or Docker)*	The application is developed and tested locally using Streamlit and can be deployed to cloud platforms with minimal configuration in future versions.
5	**Version Control & CI/CD**(e.g., GitHub, GitLab, Jenkins)	**Git & GitHub**	Enables version control, collaborative development, code review, issue tracking, and future CI/CD integration for team-based development.
6	**Third-Party APIs / Other Tools**(e.g., Stripe, Twilio, Google Maps)	**OpenAI Whisper, Sentence-Transformers (Sentence-BERT), Librosa, PyTorch, SoundFile, Matplotlib, ReportLab, NLTK**	Whisper performs speech-to-text transcription, Sentence-BERT evaluates semantic similarity, Librosa extracts speech features, PyTorch supports AI models, SoundFile processes audio, Matplotlib generates waveform visualizations, ReportLab creates PDF reports, and NLTK assists with text processing.